

ReGo - Visual Diagram References

Use these as templates to create actual diagrams

1. Database ER Diagram Reference

Tool Recommendations:

- [draw.io](#) (free, web-based)
- [dbdiagram.io](#) (specifically for database diagrams)
- [Lucidchart](#)
- [Microsoft Visio](#)

How to create:

1. Go to [dbdiagram.io](#)
2. Copy the code below
3. Paste and it will generate the diagram automatically

```
dbdiagram
```

```
Table users {
  user_id uuid [pk]
  email varchar [unique, not null]
  password_hash varchar [not null]
  first_name varchar [not null]
  last_name varchar [not null]
  phone varchar
  role varchar [not null]
  department varchar
  employee_id varchar [unique]
  reporting_manager_id uuid [ref: > users.user_id]
  is_active boolean [default: true]
  created_at timestamp [default: `now()`]
  updated_at timestamp [default: `now()`]
}
```

```
Table travel_requests {
  request_id uuid [pk]
  user_id uuid [ref: > users.user_id, not null]
  request_number varchar [unique, not null]
  travel_type varchar [not null]
  purpose text [not null]
  destination_country varchar [not null]
  destination_city varchar [not null]
  departure_date date [not null]
  return_date date [not null]
  estimated_cost decimal
  approved_budget decimal
  actual_cost decimal
  currency varchar [default: 'INR']
  status varchar [not null, default: 'DRAFT']
  current_approver_id uuid [ref: > users.user_id]
  special_requirements text
  emergency_contact_name varchar
  emergency_contact_phone varchar
  is_urgent boolean [default: false]
  submitted_at timestamp
  approved_at timestamp
  completed_at timestamp
  created_at timestamp [default: `now()`]
  updated_at timestamp [default: `now()`]
}
```

```
Table approvals {
  approval_id uuid [pk]
  request_id uuid [ref: > travel_requests.request_id, not null]
```

```
approver_id uuid [ref: > users.user_id, not null]
approval_level integer [not null]
status varchar [not null, default: 'PENDING']
comments text
approved_budget decimal
conditions text
reviewed_at timestamp
created_at timestamp [default: `now()`]
}
```

```
Table documents {
  document_id uuid [pk]
  request_id uuid [ref: > travel_requests.request_id, not null]
  uploaded_by uuid [ref: > users.user_id, not null]
  document_type varchar [not null]
  file_name varchar [not null]
  file_size_bytes bigint
  mime_type varchar
  blob_storage_url text [not null]
  ocr_status varchar [default: 'PENDING']
  ocr_extracted_data jsonb
  ocr_confidence_score decimal
  ocr_processed_at timestamp
  verification_status varchar [default: 'PENDING']
  verified_by uuid [ref: > users.user_id]
  verification_comments text
  verified_at timestamp
  upload_attempt integer [default: 1]
  uploaded_at timestamp [default: `now()`]
  updated_at timestamp [default: `now()`]
}
```

```
Table bookings {
  booking_id uuid [pk]
  request_id uuid [ref: > travel_requests.request_id, not null]
  booked_by uuid [ref: > users.user_id, not null]
  booking_type varchar [not null]
  booking_details jsonb [not null]
  cost decimal [not null]
  currency varchar [default: 'INR']
  status varchar [default: 'CONFIRMED']
  confirmation_number varchar
  vendor_name varchar
  booking_date timestamp [default: `now()`]
  travel_date timestamp
  created_at timestamp [default: `now()`]
}
```

```
Table expenses {
  expense_id uuid [pk]
  request_id uuid [ref: > travel_requests.request_id, not null]
  submitted_by uuid [ref: > users.user_id, not null]
  expense_category varchar [not null]
  description text [not null]
  amount decimal [not null]
  currency varchar [default: 'INR']
  converted_amount decimal
  exchange_rate decimal
  expense_date date [not null]
  receipt_document_id uuid [ref: > documents.document_id]
  vendor_name varchar
  status varchar [default: 'SUBMITTED']
  approved_by uuid [ref: > users.user_id]
  approved_amount decimal
  approval_comments text
  approved_at timestamp
  reimbursement_status varchar [default: 'PENDING']
  reimbursed_amount decimal
  reimbursement_date date
  transaction_reference varchar
  submitted_at timestamp [default: `now()`]
  created_at timestamp [default: `now()`]
}
```

```
Table notifications {
  notification_id uuid [pk]
  user_id uuid [ref: > users.user_id, not null]
  notification_type varchar [not null]
  title varchar [not null]
  message text [not null]
  related_entity_type varchar
  related_entity_id uuid
  is_read boolean [default: false]
  read_at timestamp
  delivery_method varchar
  delivery_status varchar [default: 'PENDING']
  sent_at timestamp
  created_at timestamp [default: `now()`]
}
```

```
Table ai_recommendations {
  recommendation_id uuid [pk]
  request_id uuid [ref: > travel_requests.request_id, not null]
  recommendation_type varchar [not null]
```

```
recommendation_data jsonb [not null]
confidence_score decimal
was_selected boolean [default: false]
selected_option_index integer
user_feedback text
generated_at timestamp [default: `now()`]
expires_at timestamp
}
```

```
Table audit_logs {
  log_id uuid [pk]
  user_id uuid [ref: > users.user_id]
  action varchar [not null]
  entity_type varchar [not null]
  entity_id uuid [not null]
  old_values jsonb
  new_values jsonb
  ip_address varchar
  user_agent text
  created_at timestamp [default: `now()`]
}
```

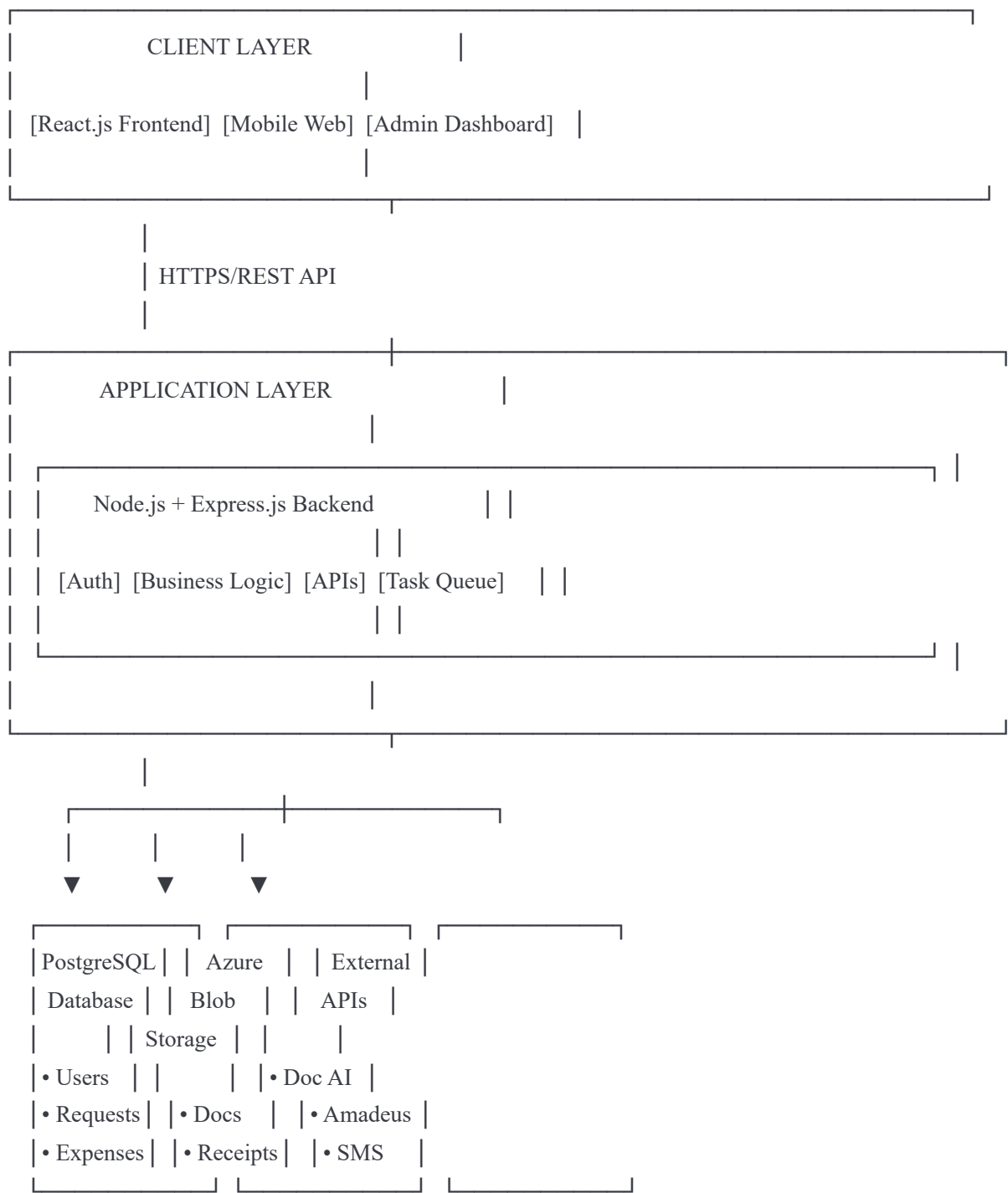
2. System Architecture Diagram Reference

Tool: draw.io (free, web-based)

Instructions:

1. Go to app.diagrams.net
2. Create new diagram
3. Use rectangles for components, arrows for connections
4. Follow this structure:

Layout (Top to Bottom):



Color Coding:

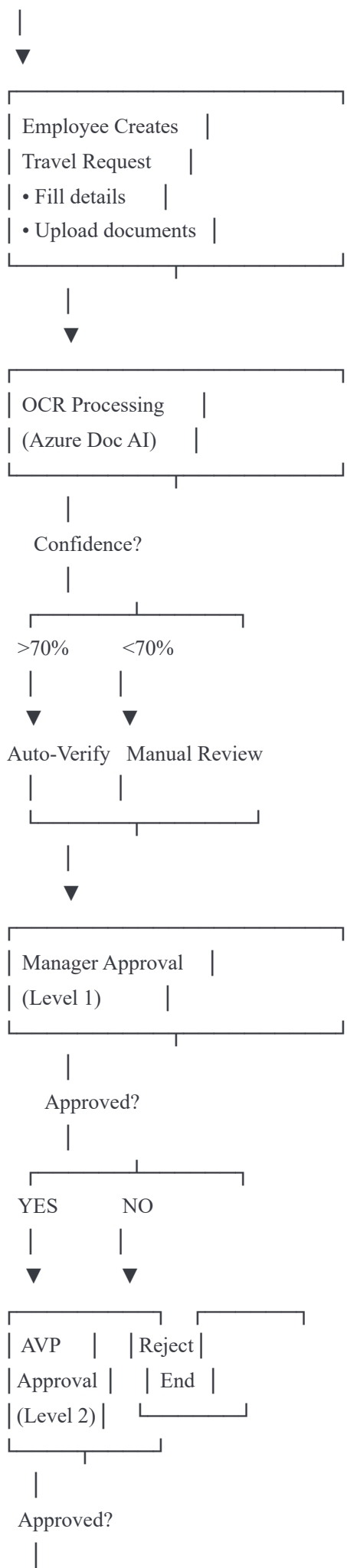
- Client Layer: Light Blue (#E3F2FD)
- Application Layer: Light Green (#E8F5E9)
- Data Layer: Light Orange (#FFF3E0)
- External APIs: Light Purple (#F3E5F5)

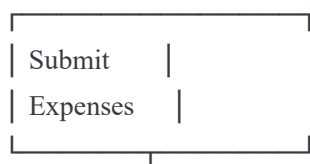
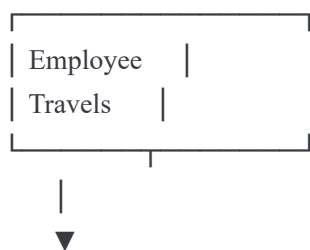
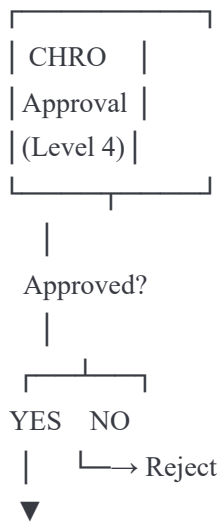
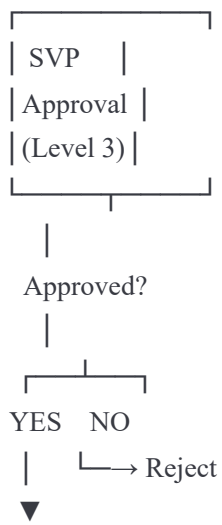
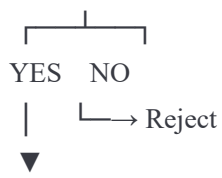
3. Travel Request Workflow Diagram

Tool: draw.io or Lucidchart

Layout (Vertical Flow):

Start





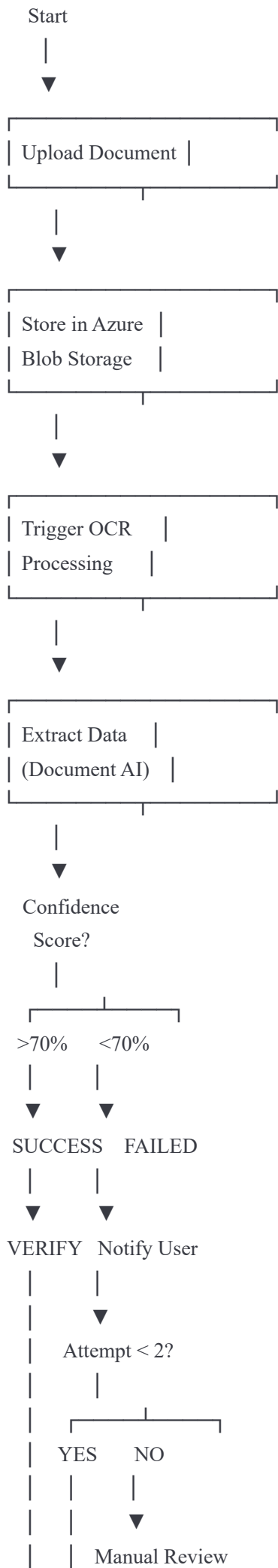


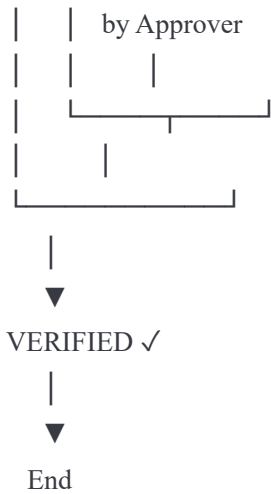
Shapes:

- Rectangles: Process steps
- Diamonds: Decision points
- Circles: Start/End
- Arrows: Flow direction

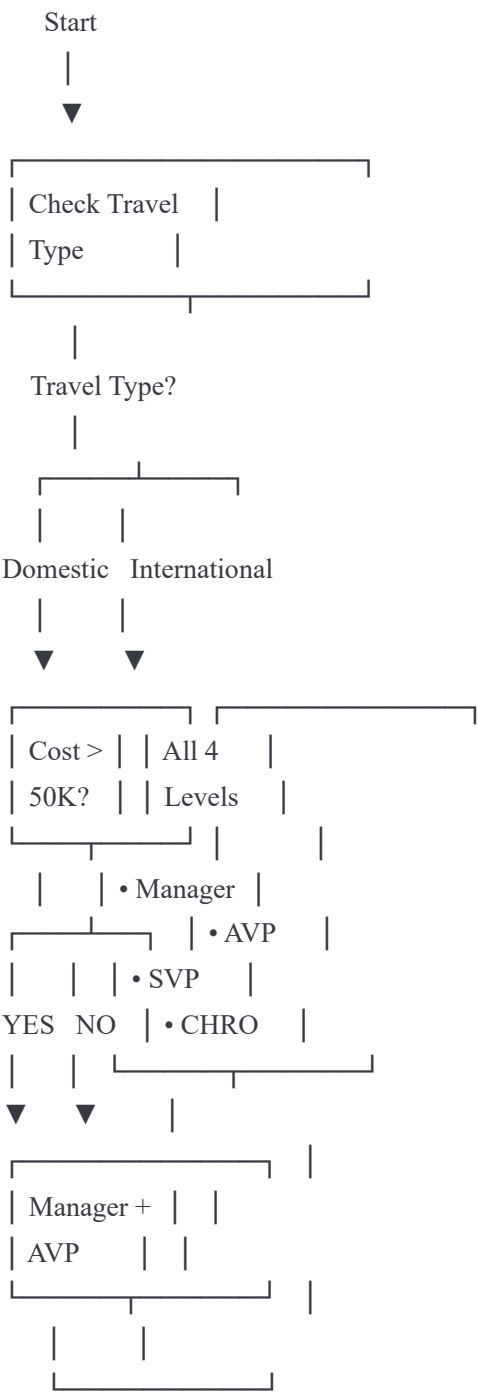
4. Document Verification Workflow

Tool: draw.io





5. Approval Chain Decision Logic Flowchart

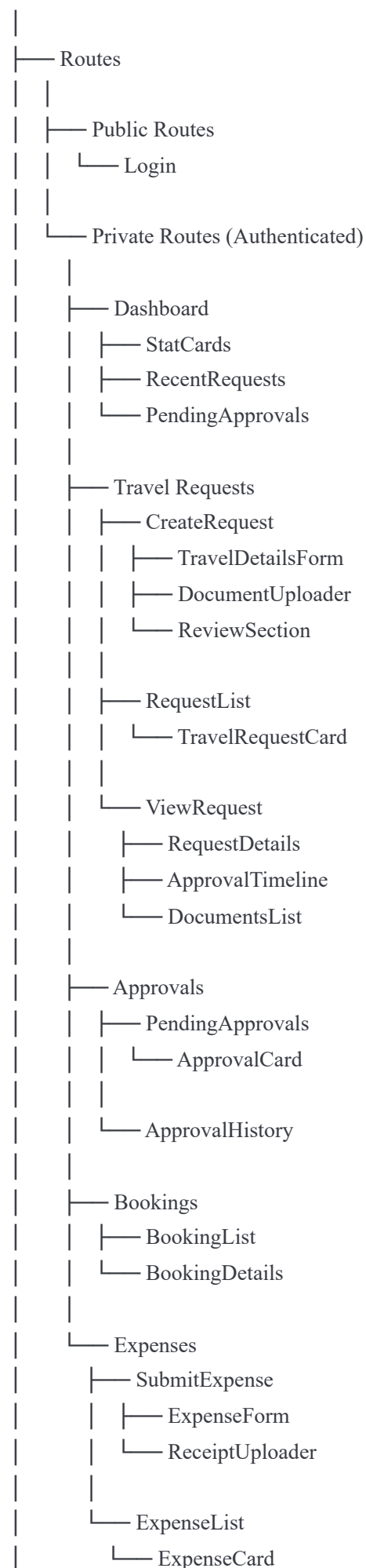




6. Frontend Component Hierarchy

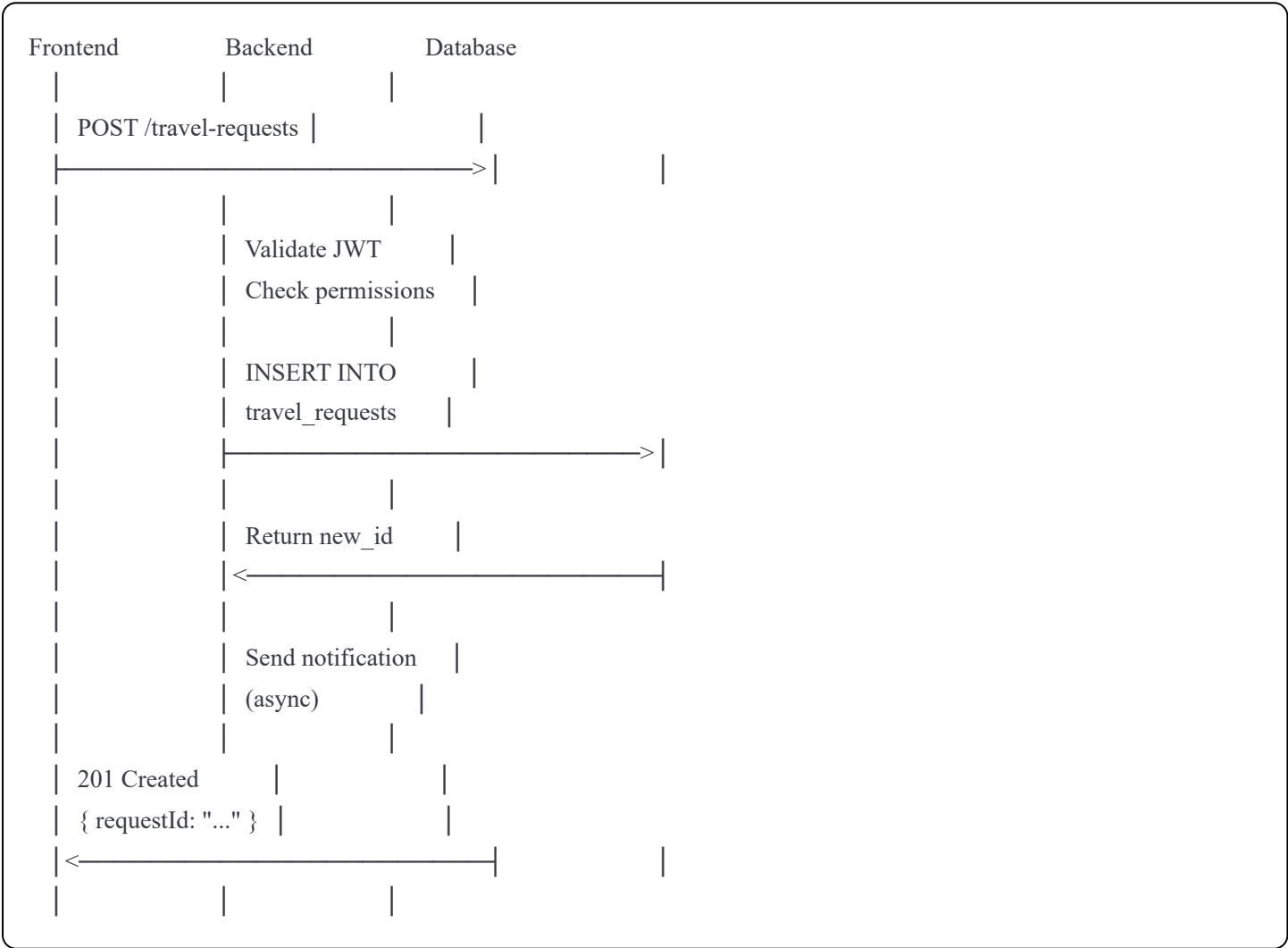
Tool: draw.io or any diagramming tool

App

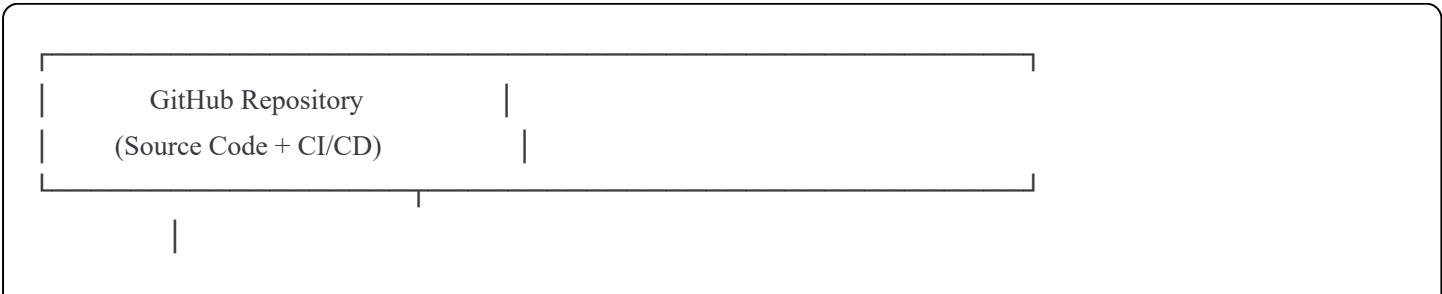


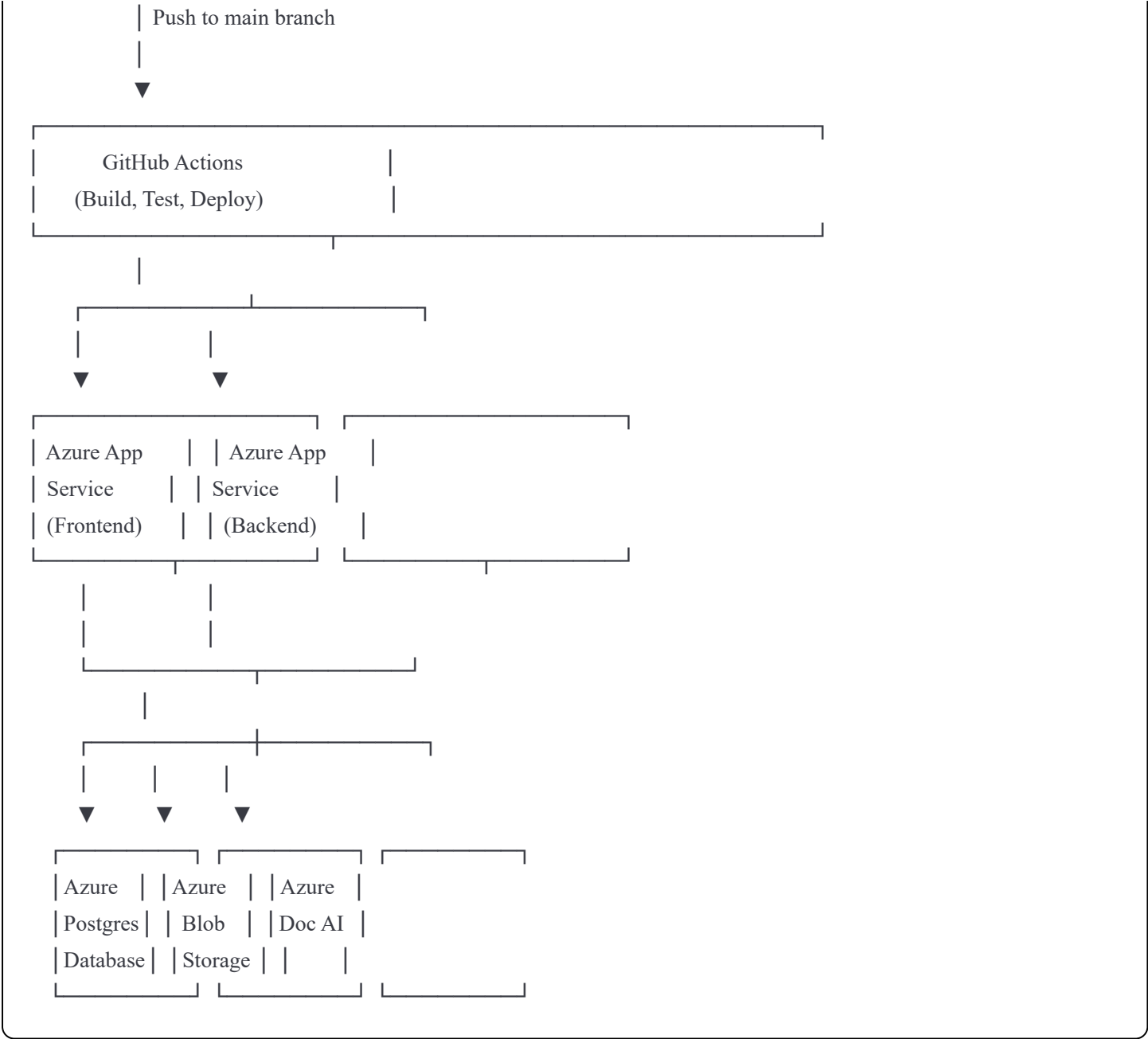


7. API Request/Response Flow Diagram



8. Deployment Architecture





9. Week-by-Week Progress Timeline

Tool: Excel, Google Sheets, or Project Management tool

Week	Milestone	Status	Demo
1	Foundation & Auth	In Progress	Login system
2	Travel Requests & Approvals	Pending	Full workflow
3	AI & OCR Integration	Pending	Document verification
4	Bookings & Expenses	Pending	Expense tracking
5	Testing & Deployment	Pending	Production app

Gantt Chart Structure:

- Week 1: Days 1-7 (Project setup, Database, Auth, Basic UI)
- Week 2: Days 8-14 (Travel module, Document upload, Approvals)

- Week 3: Days 15-21 (OCR, AI recommendations, Manual verification)
 - Week 4: Days 22-28 (Bookings, Expenses, Notifications)
 - Week 5: Days 29-35 (Testing, Fixes, Polish, Deploy)
-

10. User Journey Map

Tool: draw.io, Miro, or Figma

EMPLOYEE JOURNEY

Phase 1: Request Creation

1. Login to ReGo	
2. Click "New Travel Request"	
3. Fill travel details	
4. Upload passport/documents	
5. Submit for approval	

Phase 2: Waiting for Approval

1. Receive email notification	
2. Check status in dashboard	
3. View approval progress	
4. Receive approval confirmation	

Phase 3: Travel Preparation

1. View booking details	
2. Download flight tickets	
3. Check hotel confirmation	
4. Review visa status	

Phase 4: Post-Travel

1. Upload receipts	
2. Submit expense claims	
3. Track reimbursement status	
4. Receive payment notification	

APPROVER JOURNEY

1. Receive approval notification	
2. Login and view pending approvals	
3. Review travel request details	
4. Check uploaded documents	
5. Verify budget and policy	

| 6. Approve or reject with comments |

| 7. Notification sent to next approver |

How to Use These References

For Database Diagram:

1. Go to **dbdiagram.io**
2. Copy the code from section 1
3. Paste in the editor
4. The diagram will auto-generate
5. Export as PNG or PDF

For Architecture Diagrams:

1. Go to **app.diagrams.net** (draw.io)
2. Create new diagram
3. Use these shapes:
 - **Rectangle**: Components
 - **Cylinder**: Databases
 - **Cloud**: Cloud services
 - **Arrows**: Data flow
4. Follow the ASCII layouts above
5. Add colors as suggested

For Flowcharts:

1. Use **draw.io** or **Lucidchart**
2. Follow the flow from top to bottom
3. Use these shapes:
 - **Rounded Rectangle**: Start/End
 - **Rectangle**: Process
 - **Diamond**: Decision
 - **Arrow**: Flow direction

For Timeline/Gantt:

1. Use **Excel**, **Google Sheets**, or **Microsoft Project**

2. Create table with: Week, Tasks, Start Date, End Date, Status
 3. Use conditional formatting for visual progress
-

Quick Tips

Color Scheme for Professional Look:

- Primary:  #1976D2 (Blue)
- Secondary:  #388E3C (Green)
- Accent:  #F57C00 (Orange)
- Error:  #D32F2F (Red)
- Success:  #4CAF50 (Green)
- Warning:  #FFA726 (Amber)

Font Recommendations:

- Headings: Roboto Bold or Inter Bold
- Body Text: Roboto Regular or Inter Regular
- Code: Fira Code or JetBrains Mono

Diagram Best Practices:

- Keep it simple and clean
 - Use consistent shapes for similar items
 - Add legends for colors/symbols
 - Use arrows to show direction
 - Include title and date
 - Export in high resolution (PNG 300dpi)
-

All diagrams should be created before Week 1 Demo!